

RQ3 — AI Test Refinement Evaluation on Controlled Test Websites

Phase 2 of the experiment uses five purpose-built test websites that enforce real input validation, eliminating the oracle-defeat conditions identified in Phase 1. Each website is served locally via Live Server at `http://127.0.0.1:5500/test_websites/` and its source lives in `test_websites/` in this repository.

3.1 Phase 2 — Controlled Test Websites

Website Specifications

Site 1 — Contact Form (`site1_contact`)

URL: `http://localhost:5500/test_websites/site1_contact/index.html`

Path: `test_websites/site1_contact/`

Field	Type	Constraint	Validation Rule
<code>name</code>	text	required	length 2–50
<code>email</code>	email	required	regex <code>^[^@]+@[^\@]+\.[^\@]+</code>
<code>subject</code>	select	required	one of: General Inquiry, Support, Billing, Other
<code>message</code>	textarea	required	length 10–500

Submit button: "Submit"

On success: redirects to `success.html` (URL change = oracle-detectable)

On failure: inline `.error-msg` divs populated with red text

Expected test outcomes:

Test Class	Example Input	Expected Result
BVA valid	<code>name="AI"</code> (exactly 2 chars)	PASS — form submits
BVA invalid	<code>name="A"</code> (1 char)	FAIL — error displayed
ECP invalid	<code>email="notanemail"</code>	FAIL — error displayed
ECP valid	<code>email="user@test.com"</code>	PASS (with other fields valid)
BVA boundary	<code>message = 9 chars</code>	FAIL
BVA boundary	<code>message = 10 chars</code>	PASS

Site 2 — Hotel Booking (`site2_booking`)

URL: `http://localhost:5500/test_websites/site2_booking/index.html`

Path: `test_websites/site2_booking/`

Field	Type	Constraint	Validation Rule
<code>checkin</code>	date	required	<code>≥ today</code>

Field	Type	Constraint	Validation Rule
checkout	date	required	> checkin
guests	number	required	integer 1–10
room	select	required	one of: Standard, Deluxe, Suite, Penthouse
email	email	required	regex

On success: redirects to `confirmed.html?ref=BKXXXXXX`

Cross-field rule: checkout > checkin (both must be present)

Expected test outcomes:

Test Class	Example	Expected Result
BVA valid	guests=1	PASS
BVA valid	guests=10	PASS
BVA invalid	guests=0	FAIL
BVA invalid	guests=11	FAIL
Cross-field invalid	checkout ≤ checkin	FAIL
Date past	checkin = yesterday	FAIL

Site 3 — User Registration (`site3_register`)

URL: `http://localhost:5500/test_websites/site3_register/index.html`

Path: `test_websites/site3_register/`

Field	Type	Constraint	Validation Rule
username	text	required	3–20 chars, <code>^[a-zA-Z0-9_]+\$</code>
email	email	required	regex
password	password	required	≥8 chars, ≥1 uppercase, ≥1 digit, ≥1 special char
confirmPassword	password	required	must match <code>password</code>
age	number	required	integer 18–120

On success: inline success banner: "☒ Account created! Welcome, {username}"

Expected test outcomes:

Test Class	Example	Expected Result
BVA valid	username="abc" (3 chars)	PASS
BVA invalid	username="ab" (2 chars)	FAIL
ECP invalid	password="password" (no digit/special/upper)	FAIL
ECP valid	password="Test@123"	PASS
Cross-field	confirmPassword ≠ password	FAIL
BVA boundary	age=17	FAIL
BVA boundary	age=18	PASS

Site 4 — Product Search ([site4_search](#))

URL: http://localhost:5500/test_websites/site4_search/index.html

Path: [test_websites/site4_search/](#)

Field	Type	Constraint	Validation Rule
keywords	text	required	length ≥ 2
category	select	required	one of: Electronics, Clothing, Books, Home & Garden, Sports
minPrice	number	optional	≥ 0
maxPrice	number	optional	$\geq \text{minPrice}$, ≤ 10000

On success: reveals [#results](#) section with 3 dummy product cards

Cross-field rule: $\text{maxPrice} \geq \text{minPrice}$

Expected test outcomes:

Test Class	Example	Expected Result
BVA valid	keywords="ab" (2 chars)	PASS
BVA invalid	keywords="a" (1 char)	FAIL
Cross-field invalid	minPrice=500, maxPrice=100	FAIL
ECP valid	minPrice=0, maxPrice=10000	PASS
ECP invalid	maxPrice=10001	FAIL

Site 5 — Feedback Survey ([site5_feedback](#))

URL: http://localhost:5500/test_websites/site5_feedback/index.html

Path: [test_websites/site5_feedback/](#)

Field	Type	Constraint	Validation Rule
name	text	required	non-empty
email	email	optional	if provided: regex
rating	radio (1–5)	required	one selected
categories	checkboxes	required	≥ 1 checked
comment	textarea	optional	length ≤ 300
phone	text	optional	if provided: 7–15 digits

On success: redirects to [thank-you.html](#)

Expected test outcomes:

Test Class	Example	Expected Result
Required field	name=""	FAIL
Optional valid	email=""	PASS (if other fields valid)
Optional invalid	email="notvalid"	FAIL

Test Class	Example	Expected Result
BVA invalid	comment = 301 chars	FAIL
BVA valid	comment = 300 chars	PASS
Required group	no rating selected	FAIL
Required group	no category checked	FAIL
Optional valid	phone=""	PASS
Optional BVA	phone = 6 digits	FAIL

3.2 Phase 2 Results

Updated after each pipeline run. Run: `python scripts/run_test_websites.py --site <0-4>`

Site	Pages	Tests Before	Tests After	Δ	AI Enhanced	Passed	Failed	Pass %	LLM Calls	Cost (\$)	Stop
site1_contact	1	52	58	+6	58	46	12	79.3%	4	0.008	completed
site2_booking	1	41	46	+5	46	38	8	82.6%	5	0.010	completed
site3_register	1	51	56	+5	56	27	29	48.2%	7	0.014	completed
site4_search	1	32	38	+6	38	25	13	65.8%	1	0.002	completed
site5_feedback	1	116	117	+1	97	88	29	75.2%	6	0.012	completed
TOTAL	5	292	315	+23	295	224	91	71.1%	23	0.046	